Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack Software Engineer with 2+ years shipping features end to end in **TypeScript**, **React Native**, React, and **Java**, with real-time WebSocket systems, payment and refund flows, SQL data models, and AWS deployments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed multi-tenant **SQL** schemas with row-level security for 500+ users, keeping each tenant's data isolated and tuning indexed queries to cut dashboard latency from 1.2s to 350ms.
- Restructured fault-tolerant asynchronous job processing with explicit state transitions and failure isolation, handling 1K+ jobs at 3x **throughput** and shipping integrations through CI/CD at 99%+ reliability across 5K+ monthly jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- **Owned** upload, rubric, grading, and history features from design through production in a React app with REST APIs and MySQL on AWS, serving 150+ instructors and 10K+ daily requests.
- Integrated GPT-4o, Claude, and Qwen-VL with OCR fallbacks and multithreaded batch processing, keeping the UI responsive at peak load and cutting instructor feedback turnaround 60% at 90%+ grading consistency.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built object-oriented **Java** and Python microservice modules with Singleton and Factory patterns, covered by 20+ automated unit and integration suites run nightly in Jenkins across Windows, Linux, and macOS.
- Optimized API integrations with pagination, field filtering, and caching, improving response times 30% for low-bandwidth users, and redesigned AWS Glue and S3 pipelines for 68% higher efficiency.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **TypeScript** React and Java Spring Boot platform with **real-time WebSocket** updates and conflict resolution for shared resources, syncing 2M+ records across multiple locations with consistent state.
- Built authentication, role-based access, billing, and **notification** services with SMS and email alerts, shipped through Jenkins CI/CD with Jest and Selenium tests and Docker staging releases, cutting deployment time 50%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built a **React Native** Android and iOS app backend with Firebase real-time sync and push notifications, improving booking confirmation speed 40% and cross-platform compatibility 15% across Android, iOS, and web.
- Built PCI-compliant multi-currency **payment** infrastructure for cards, wallets, and PayPal with idempotent webhooks for refunds, disputes, and chargebacks, cutting manual refund handling about 70% across 50K+ booking records.

Skills

Languages: TypeScript, JavaScript, Java, Python, SQL, C++, HTML, CSS
Mobile & Frontend: React Native, React, Next.js, TailwindCSS, Responsive Design, Figma
Backend & Real-Time: Spring Boot, Node.js, Express.js, REST APIs, WebSockets, Firebase, Microservices
Data & Payments: PostgreSQL, MySQL, MongoDB, Stripe, PayPal, Webhooks, JWT, Role-Based Access
Cloud & Practices: AWS, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Jest, Selenium, Git, Agile, Linux

Projects

Real-Time Multiplayer Card Table | *TypeScript, Node.js, WebSockets, Redis, React Native*

- Designed a **server-authoritative** card-game engine where a TypeScript service owns shuffling, turns, and pot math, syncs table state to React Native clients over WebSockets, and recovers **reconnects**.

Wallet Deposits with KYC Checks | *Java, Spring Boot, PostgreSQL, Stripe, AWS*

- Designed a Spring Boot wallet service with a double-entry ledger, **KYC** verification before first withdrawal, geolocation checks by state, and AML rules that flag unusual **deposit** patterns.

FinNexus Live | *Next.js, TypeScript, Plaid, MongoDB, WebSockets*

- Built a **fintech** platform with Plaid bank linking, a token-based firewall, and real-time WebSocket updates, supporting 50,000+ users and 1M+ transactions with secure, filtered access and Kubernetes-based CI/CD deployments.